

DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

BIOLOGY — CLASS 6 EXAMINATION

SUBJECT: BIOLOGY

TIME ALLOWED: 1.5 HOURS

MAX MARKS: 35

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. All questions are compulsory.
2. Read all instructions carefully before writing.
3. Marks are indicated next to each question.
4. For numerical problems, clearly show your formula, substitution, and final answer with correct units.

SECTION A — MULTIPLE CHOICE QUESTIONS

Select the most appropriate option from the choices given below.

Q1. A student adds a few drops of dilute iodine solution to a peeled potato slice. The potato slice turns blue-black. This food test confirms the presence of which nutrient? (1)

- (a) a) Proteins (b) b) Fats
(c) c) Starch (d) d) Vitamin C

Q2. Which of the following adaptations helps a cactus plant survive in dry, desert environments with minimal water loss? (1)

- (a) a) Broad and thin leaves with numerous stomata (b) b) Leaves modified into sharp spines and a fleshy green stem
(c) c) Soft, delicate stem with a shallow root system (d) d) Woody, tall trunk with branches spread high in the air

Q3. Which type of joint in the human body allows movement in all directions, similar to the movement of a computer joystick? (1)

- (a) a) Hinge joint (b) b) Pivotal joint
(c) c) Ball and socket joint (d) d) Fixed joint

Q4. An earthworm does not have specialized lungs or gills. Through which of the following organs does it exchange respiratory gases? (1)

- (a) a) Moist skin (b) b) Tracheae
(c) c) Spiracles (d) d) Scales

SECTION B — SHORT QUESTIONS

Answer the following questions in 2 to 4 sentences.

Q1. Define 'Biotic' and 'Abiotic' components of a habitat, and provide one example for each. (2)

Q2. Why are vitamins and minerals collectively referred to as 'protective foods'? (2)

Q3. Distinguish between a taproot system and a fibrous root system based on their structural appearance. (2)

SECTION C — NUMERICAL PROBLEMS

Solve the following quantitative biology problems. Show all formula steps, substitutions, and write your final answer with the correct units.

- Q1. A Class 6 student consumes a breakfast that contains 20 grams of carbohydrates, 10 grams of proteins, and 5 grams of fats. Knowing that 1 gram of carbohydrate provides 4 kcal of energy, 1 gram of protein provides 4 kcal, and 1 gram of fat provides 9 kcal, calculate the total energy (in kcal) obtained by the student from this breakfast. (5)
- Q2. During physical education, Rohan's resting heart rate is recorded as 72 beats per minute (bpm). After running around the playfield, Rohan measures his pulse rate by counting 27 heartbeats in exactly 15 seconds. Calculate Rohan's post-exercise heart rate in beats per minute (bpm), and find the total increase in his heart rate compared to his resting rate. (5)
- Q3. In a laboratory experiment on transpiration, a potted plant lost 150 ml of water over a 6-hour period. Calculate the average rate of transpiration of this plant in milliliters per hour (ml/hr). If the plant has exactly 50 leaves, calculate the average amount of water lost by a single leaf per hour, assuming all leaves transpire at an equal rate. (5)
- Q4. A student uses a square quadrat of area 2 square meters (2 m^2) to estimate the population of wild dandelion plants on a school lawn. In one throw, she counts 12 dandelions inside the quadrat. Calculate the population density of dandelions (plants per square meter). Using this density, estimate the total expected population of dandelions in a large garden plot measuring 50 square meters (50 m^2). (5)
- Q5. A 250-gram commercial pack of infant cereal is tested in a lab. The analysis reveals that it contains 175 grams of carbohydrates and 25 grams of proteins, with the remainder composed of water and dietary minerals. Calculate the percentage of carbohydrates and the percentage of proteins present in this cereal pack. (5)

ANSWER KEY

SECTION A — MULTIPLE CHOICE QUESTIONS

- Q1. c) Starch [1 mark]
- Q2. b) Leaves modified into sharp spines and a fleshy green stem [1 mark]
- Q3. c) Ball and socket joint [1 mark]
- Q4. a) Moist skin [1 mark]

SECTION B — SHORT QUESTIONS

- Q1. Biotic components refer to all the living organisms present in an ecosystem or habitat, such as plants, animals, and microorganisms. Abiotic components, on the other hand, are the non-living physical and chemical elements in the environment, such as soil, water, air, and sunlight. [2 marks]
- Q2. Vitamins and minerals are called protective foods because they defend our body against various infections and diseases. Although needed in small quantities, they are essential for maintaining healthy eyes, bones, teeth, gums, and for the smooth functioning of metabolic processes. [2 marks]
- Q3. A taproot system consists of a single, thick main root (primary root) that grows vertically down into the soil with smaller lateral branch roots (e.g., carrot). In contrast, a fibrous root system consists of a cluster of thin, similarly sized roots arising directly from the base of the stem without any main primary root (e.g., grass). [2 marks]

SECTION C — NUMERICAL PROBLEMS

- Q1. Formula: [5 marks]
$$\text{Total Energy} = (\text{Mass of Carbohydrates} \times \text{Energy Value}) + (\text{Mass of Proteins} \times \text{Energy Value}) + (\text{Mass of Fats} \times \text{Energy Value})$$

Substitution:
$$\text{Total Energy} = (20 \text{ g} \times 4 \text{ kcal/g}) + (10 \text{ g} \times 4 \text{ kcal/g}) + (5 \text{ g} \times 9 \text{ kcal/g})$$
$$\text{Total Energy} = 80 \text{ kcal} + 40 \text{ kcal} + 45 \text{ kcal}$$
$$\text{Total Energy} = 165 \text{ kcal}$$

Final Answer: 165 kcal
- Q2. Formula: [5 marks]
1. Post-Exercise Heart Rate (bpm) = (Number of beats in 15 seconds) * 4
2. Increase in Heart Rate = Post-Exercise Heart Rate - Resting Heart Rate

Substitution:
1. Post-Exercise Heart Rate = 27 beats * 4 = 108 bpm
2. Increase in Heart Rate = 108 bpm - 72 bpm = 36 bpm

Final Answer:
Rohan's post-exercise heart rate is 108 bpm. The increase in his heart rate is 36 bpm.
- Q3. Formula: [5 marks]
1. Rate of Transpiration = Total Water Lost / Total Time
2. Water Lost per Leaf per Hour = Rate of Transpiration / Total Number of Leaves

Substitution:
1. Rate of Transpiration = 150 ml / 6 hours = 25 ml/hr
2. Water Lost per Leaf per Hour = 25 ml/hr / 50 leaves = 0.5 ml/leaf/hr

Final Answer:
The average rate of transpiration is 25 ml/hr. The average water lost per leaf per hour is 0.5 ml/leaf/hr.

Q4. Formula: [5 marks]
1. Population Density = Total Number of Plants counted / Area of the Quadrat
2. Estimated Total Population = Population Density * Area of Garden Plot

Substitution:
1. Population Density = 12 plants / 2 m² = 6 plants/m²
2. Estimated Total Population = 6 plants/m² * 50 m² = 300 plants

Final Answer:
The population density of dandelions is 6 plants/m². The estimated total population in the 50 m² garden plot is 300 plants.

Q5. Formula: [5 marks]
1. Percentage of Carbohydrates = (Mass of Carbohydrates / Total Mass of Cereal) * 100
2. Percentage of Proteins = (Mass of Proteins / Total Mass of Cereal) * 100

Substitution:
1. Percentage of Carbohydrates = (175 g / 250 g) * 100 = 0.7 * 100 = 70%
2. Percentage of Proteins = (25 g / 250 g) * 100 = 0.1 * 100 = 10%

Final Answer:
The cereal contains 70% carbohydrates and 10% proteins.
